

SIMRAN SAINAND REVANKAR

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Research Experience

Project Associate – I

Department of Electrical Engineering

Indian Institute of Technology Palakkad, India

Mar 2025 – Jun 2026

Project: Stochastic Resonance Based Detectors Using Thin-Film Integrated Ferroelectrics

Funding: Core Research Grant, Science and Engineering Research Board (SERB)

Project duration: Dec 2022 – Dec 2025

- Developed reproducible **PVDF-TrFE thin-film fabrication** workflows using spin coating, Langmuir–Blodgett deposition and drop casting.
- Investigated the influence of **solution chemistry, solvent composition, drying and annealing conditions** on thin-film properties and electrical behaviour.
- Fabricated **ferroelectric capacitor devices** and investigated structure-property relationships through electrical measurements.
- Performed **P–E hysteresis and PUND measurements** using a Radiant Technologies ferroelectric tester.
- Gained exposure to thin-film deposition workflows including sputtering and thermal evaporation within device-fabrication processes.
- Analysed experimental datasets and investigated relationships between processing conditions, thin-film properties and device performance.

Independent Researcher

Department of Chemistry

JSS Science & Technology University, India

Nov 2024 – Mar 2025

Perovskite materials, interfaces and functional polymer systems

- Investigated Cu-based and Rb-based perovskite systems for energy-harvesting applications.
- Examined how **composition and nanoscale interfaces** influence material and optoelectronic behaviour.

Graduate Researcher

Department of Chemistry

JSS Science & Technology University, India

Dec 2023 – Sep 2024

Halide perovskites, polymer nanocomposites and functional devices

- Synthesized and characterized **halide perovskite materials**, including lead-based and lead-free compositions, and developed perovskite/polymer nanocomposites for functional devices.
- Investigated relationships between **material composition, morphology, interfaces and device performance**.
- Fabricated flexible triboelectric devices and evaluated their electrical response under different material compositions and processing conditions.
- Characterized materials using **XRD, DSC, SEM, AFM, UV-Vis, FTIR and contact-angle measurements data interpretation**.
- Published research on fluorescent perovskite/polymer nanocomposites for self-powered UV sensing.

Research Intern

Department of Plantation Product, Spices and Flavour Technology

CSIR–Central Food Technological Research Institute, India

Nov 2023 – Dec 2023

- Optimized curcumin content in oleoresin, extraction was done using spectroscopic and chromatographic analysis.
- Performed characterization using **1D NMR, FTIR, UV-Vis and HPLC**.

Research Expertise

Materials and processing

Halide perovskites; lead-based and lead-free compositions; PVDF-TrFE, polyurethane, EVA, PMMA; perovskite/polymer nanocomposites; solution preparation; flexible functional materials.

Characterisation

XRD; FTIR; Raman; contact angle; SEM; AFM; HPLC (data analysis); | UV-Vis and 1D NMR (hands-on experience)

Device research

Triboelectric nanogenerators; self-powered UV sensing; electrode/device assembly; electrical-response evaluation.

Research data

Python; OriginPro; ImageJ; Gwyddion; quantitative analysis; scientific visualisation; research documentation.

Publications

- S. S. Revankar, Y. V. Lokesh, A. S. Manjunatha, . . . , B. S. Madhukar, “Fluorescent CsPbBr₃@Cs₄PbBr₆/PU polymer nanocomposite-based triboelectric nanogenerator for self-powered UV sensing,” *Materials Today Chemistry*, 44 (2025), 102601. **First author**. DOI: 10.1016/j.mtchem.2025.102601
- N. Manjushree, Y. V. Lokesh, K. R., . . . , **S. S. Revankar**, . . . , B. S. Madhukar, “Characterization and DFT investigation of CuWO₄@CMC/PVA composite with down conversion ability,” *Journal of Molecular Structure*, 1325 (2025), 140780. **7th author**. DOI: 10.1016/j.molstruc.2024.140780
- K. M. Gowda, Y. V. Lokesh, U. G. P. . . . , **S. S. Revankar**, . . . , B. S. Madhukar, “Flexible, lead-free Cs₃Bi₂Br₉@EVA nanocomposite triboelectric nanogenerator for energy harvesting and tactile sens-

ing,"*Emergent Materials*, 8 (2025), 5827–5842. **7th author.** DOI: 10.1007/s42247-025-01170-0

Education

M.Sc. Chemistry

CGPA: 9.50/10 | GPA: 3.95/4.0

JSS Science & Technology University, India

2022 – 2024

Thesis: “Flexible and fluorescent CsPbBr₃@Cs₄PbBr₆/PU polymer nanocomposite as triboelectric nanogenerator for self-powered UV sensor.”

B.Sc. Chemistry, Zoology & Biotechnology

CGPA: 8.90/10 | 3.7/4.0

Maharani Science College for Women, India

2019 – 2022

Research Project: “Survey on PCOS collecting data from 104 individuals.”

- Conducted a survey-based study involving data collection from 104 individuals and performed statistical analysis of the collected data.
- Interpreted the findings through discussions with a gynaecologist and a PhD student and prepared the corresponding research report.

Advanced Research Training

- **Nanostructured Optoelectronic Materials** – NIT Warangal, India, 2025.
- **Nanoelectronics & Circuits** – IIT Palakkad, India, audited, 2025.

Research Funding & Recognition

- **KSCST Student Research Project:** “Lead-free Perovskite/Polymer Nanocomposite as Triboelectric Nanogenerator.”
Funded by the **Karnataka State Council for Science and Technology (KSCST), Government of Karnataka**; Ref. 47S_MSC_0097; Approved funding: INR 7,000.
- **Best Oral Presentation Award**, International Conference on Polymers for Sustainable Development, 2025.

Selected Presentations

- Poster Presentation – 17th International Postgraduate Summer School on Green Chemistry, Venice, Italy, 2025.
- Oral Presentation – International Conference on Nanostructured Materials and Nanocomposites, 2024.
- Poster Presentation – International Symposium on Nanomaterials for Health and Energy, 2024.
- Poster Presentation – 47th Student Project Program, State-Level Competition, 2024.

References

Dr. Arvind Ajoy

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Dr. Sachin R. Chaudhari

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